Mohammed Asim

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Education

University of California, Davis (UC Davis) — Davis, CA
Master of Science (Thesis) in Computer Science, GPA: 3.75/4.0 — *Sept. 2022 – March 2025*

Jamia Millia Islamia (Ranked 3rd among Central Universities in India) — New Delhi, India
Bachelor of Technology in Computer Engineering, GPA: 9.32/10.0, *Top 5% of the batch.* — *Aug. 2018 – June 2022*

Technical Skills

Programming Languages: Python, C++, Java, JavaScript, TypeScript
Frameworks: Node.js, React, Express.js, Next.js, REST APIs
Databases: MongoDB, PostgreSQL, Redis, SQL
Developer Tools & Cloud: Docker, Kubernetes, GitHub, AWS (ECS2, Lambda, SageMaker)
Machine Learning & Data: TensorFlow, PyTorch, Keras, Pandas, NumPy, Scikit-learn, Deep Learning, Generative AI

Experience

Graduate Student Researcher — June 2023 – Oct. 2024
UC Davis Health — *Davis, CA*

- Spearheaded the development of a high-performance Convolutional Neural Network (CNN) for MRI brain segmentation, achieving a Dice Similarity Score of 0.988, reducing manual annotation time by 70%.
- Implemented data preprocessing pipelines for a dataset of over 20,000 MRI scans, improving segmentation performance across varied head orientations by 20%.

Teaching Assistant (Machine Learning Course) — March. 2023 – June 2023
UC Davis — *Davis, CA*

- Led hands-on learning experiences for 70+ students in the Machine Learning course, boosting average engagement and comprehension scores.
- Graded and provided feedback on 50+ assignments to ensure students achieved an understanding of advanced Machine Learning and Data Analysis concepts.

Software Engineer Intern — June 2021 – July 2021
Airports Authority of India — *New Delhi, India*

- Developed a web-based application using the MERN stack to streamline inventory management, reducing reliance on paperwork by 50% and improving data accuracy across departments.
- Delivered weekly progress reports and excel mockups to stakeholders, enhancing cross-departmental communication efficiency.

Projects

Generative AI SQL Chat App | *Python, Streamlit, MySQL, SQLite, Groq API* — Live App Link | GitHub

- Built an AI-powered chatbot using the Groq API to translate natural language into SQL queries, enabling seamless interaction with SQLite and user-specified MySQL databases.
- Deployed an interactive Streamlit app with a secure UI for database configuration and real-time query execution.

Sleep Disorder Prediction using Machine Learning | *Flask, Python, Scikit-Learn, AWS EC2* — Live App Link | GitHub

- Designed a machine learning–driven web application to predict sleep disorders (Insomnia, Sleep Apnea, or None) using physiological and lifestyle inputs.
- Implemented a REST API and Flask web interface integrated with a Random Forest model, deploying the full-stack application on AWS EC2.

Personal Finance App | *MERN Stack, JWT, Vite, Render, Vercel* — Live App Link | GitHub

- Engineered a full-stack personal finance tracker with secure JWT authentication, RESTful APIs, and dynamic expense visualization using MongoDB, Express, React, and Node.js.
- Deployed the backend on Render and the frontend on Vercel, enabling real-time expense tracking with a responsive UI and protected user sessions.

Publications

1. **Scour Modeling Using Deep Neural Networks** | *ICT Express, Elsevier* — Publication Link
2. **Predicting Compressive Strength of Concrete** | *Cogent Engineering, Taylor & Francis* — Publication Link
3. **Predicting Next-Day Rainfall Using Machine Learning Techniques** | *(In Review)* — Preprint